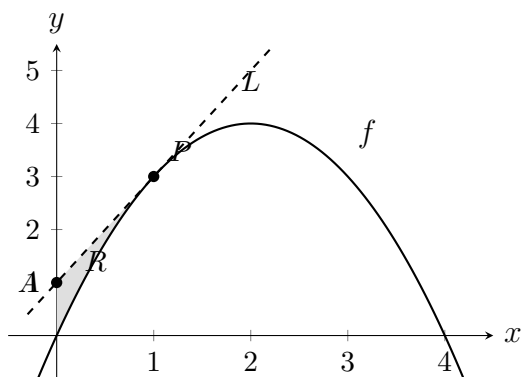


5.10 Classwork: Tangent Lines and Enclosed Area

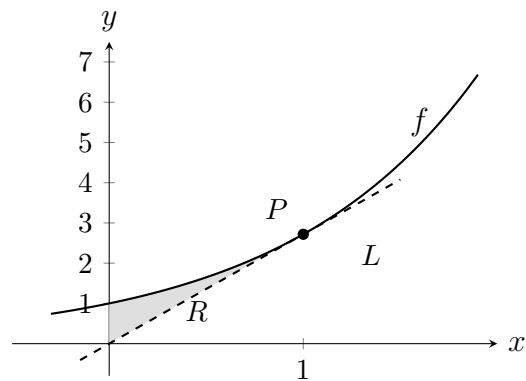
1. **(no calculator)** The graph of $f(x) = 4x - x^2$ and the line L are shown. L is tangent to the graph of f at the point $P(1, 3)$.



- (a) Show that the equation of L is $y = 2x + 1$. [2]

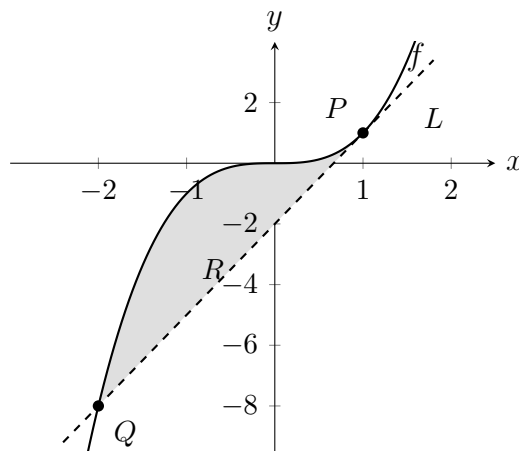
- (b) L meets the y -axis at the point A . Find the area of the shaded region R , bounded by the graph of f , the line segment AP , and the y -axis. [4]

2. **(no calculator)** The graph of $f(x) = e^x$ and the line L are shown. L is tangent to the graph of f at the point $P(1, e)$.



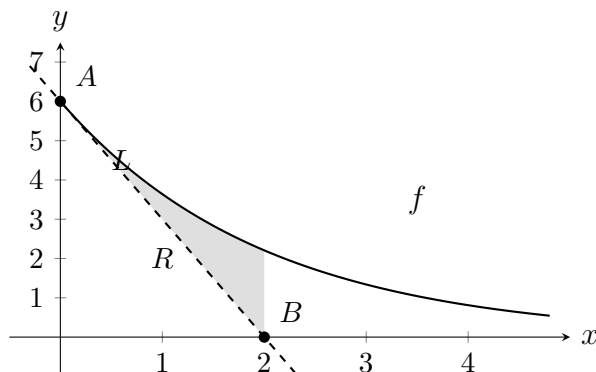
- (a) Write down $f'(x)$. [1]
- (b) Show that the equation of L is $y = ex$. [2]
- (c) Find the area of the shaded region R , bounded by the graph of f , the line L , and the y -axis. [4]

3. **(no calculator)** The graph of $f(x) = x^3$ and the line L are shown. L is tangent to the graph of f at the point $P(1, 1)$ and intersects the curve again at point Q .



- (a) Find $f'(x)$. [1]
- (b) Show that the equation of L is $y = 3x - 2$. [2]
- (c) Find the coordinates of Q . [3]
- (d) Find the area of the shaded region R , enclosed between the graph of f and the line L . [5]

4. (**calculator**) The graph of $f(x) = 6e^{-0.5x}$, for $x \geq 0$, and the line L are shown. L is tangent to the graph of f at A , where the curve meets the y -axis. B is where L crosses the x -axis.



- (a) Write down the coordinates of A . [1]
- (b) Find $f'(x)$. [2]
- (c) Show that the equation of L is $y = -3x + 6$. [2]
- (d) Find the coordinates of B . [1]
- (e) Find the area of the shaded region R , enclosed between the graph of f and the line L . [4]